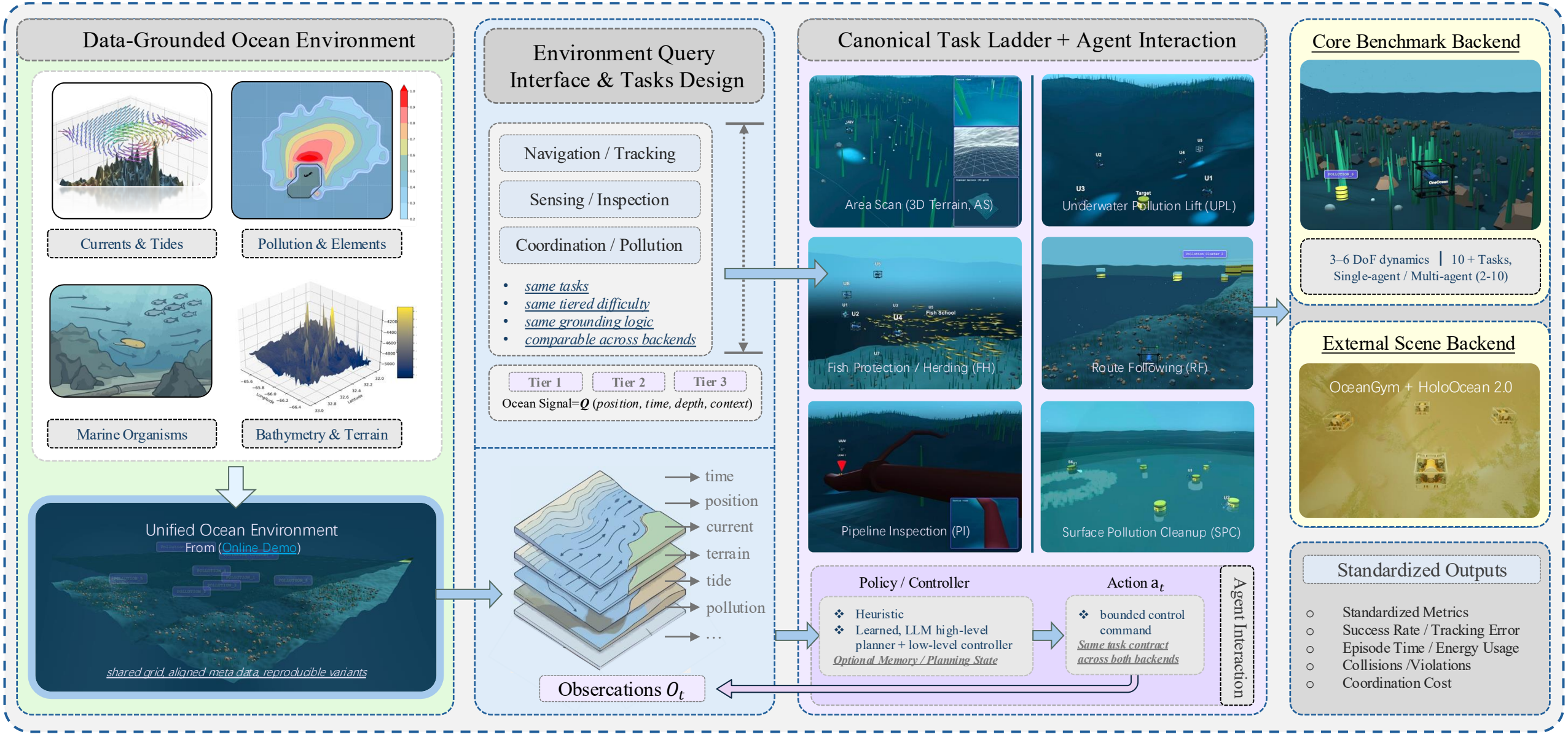
Data-Grounded Ocean Environment
Currents & Tides
Pollution & Elements
Marine Organisms
Bathymetry & Terrain
Unified Ocean Environment
From (Online Demo)
shared grid, aligned meta data, reproducible variants
Environment Query Interface & Tasks Design
Navigation / Tracking
Sensing / Inspection
Coordination / Pollution
same tasks
same tiered difficulty
same grounding logic
comparable across backends
Tier 1
Tier 2
Tier 3
Ocean Signal=Q (position, time, depth, context)
time
position
current
terrain
tide
pollution
…
Obsercations O_t
Canonical Task Ladder + Agent Interaction
Area Scan (3D Terrain, AS)
Underwater Pollution Lift (UPL)
Fish Protection / Herding (FH)
Route Following (RF)
Pipeline Inspection (PI)
Surface Pollution Cleanup (SPC)
Policy / Controller
Heuristic
Learned, LLM high-level planner + low-level controller
Optional Memory / Planning State
Action a_t
bounded control command
Same task contract across both backends
Agent Interaction
Core Benchmark Backend
3–6 DoF dynamics | 10 + Tasks, Single-agent / Multi-agent (2-10)
External Scene Backend
OceanGym + HoloOcean 2.0
Standardized Outputs
Standardized Metrics
Success Rate / Tracking Error
Episode Time / Energy Usage
Collisions /Violations
Coordination Cost